

Last name: _____
First name: _____
Matricola: _____
M.Sc. program: _____

Politecnico di Milano

Masters of Science in Biomedical Engineering; Computer Science and Engineering; Geoinformatics Engineering; High Performance Computing Engineering; Music and Acoustic Engineering; Telecommunication Engineering

FOUNDATIONS OF OPERATIONS RESEARCH

Exam of XXX XX, 2024

Time available: 2 hours and 10 minutes

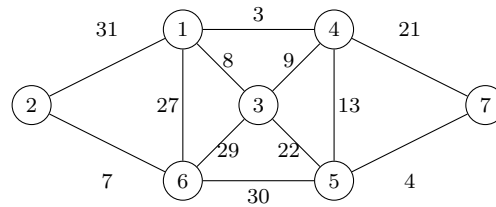
Prob. 1	Prob. 2	Prob. 3	Prob. 4	Prob. 5	Total
7	9	7.5	3.0	4.5	31

Remarks:

- Write the answers to the various questions in the specified locations **using a blue or black pen**.
- Report and explain all steps on the **back of the previous page**.
- The answers should be clear and precise.
- Calculators are not allowed. No course material or summary is allowed, except for Problem 5 which is open notes/book.

Problem 1

A company must interconnect seven new buildings via underground optical fiber cables in order to guarantee communications between every pair of buildings. Consider the following undirected graph whose nodes correspond to the buildings and whose edges correspond to the cables that can be installed between some pairs of buildings. Each edge is assigned a length expressed in multiples of 100 meters.



The goal is to determine how to interconnect all the buildings so as to minimize the total length of the cable required. For the sake of simplicity, we assume that the capacity of the cables to be installed is large enough to support any traffic between the seven buildings.

a) Provide a formal description of the graph optimization problem faced by the company, carefully specifying the inputs/outputs and precisely defining the feasible solutions.

b) Which algorithm can be used to solve the problem? Apply it to the above instance (reporting the intermediate steps on the back of the previous page) and indicate here the optimal solution found with the corresponding objective function value.

c) Consider the above optimal solution and determine for which (interval of) values of the length of the edge $\{5, 7\}$, currently equal to 400 meters, the optimal solution remains unchanged. Same question for the edge $\{3, 4\}$. In both cases, clearly explain why.

d) What is the computational complexity of the selected algorithm in terms of the number of nodes n and of edges m ? Motivate the answer **on the back of the previous page**.

Problem 2

Consider the linear programming problem:

$$\begin{array}{llllll} \min z = & -2x_1 & +2x_2 & +3x_3 & & \\ \text{s. t.} & x_1 & & +2x_3 & \leq & 2 \\ & x_1 & +x_2 & & \leq & 10 \\ & x_1, & x_2, & x_3 & \geq & 0 \end{array}$$

a) Write the problem in standard form and indicate a basic feasible solution associated with it.

b) Apply **one iteration** of the simplex method with Bland's rule to this linear program, starting from the solution $\underline{x} = (0, 0, 0)^T$. Indicate **here** the basic feasible solution found and its objective function value. Report all the steps, including the initial and the resulting tableaux, on the **back of the previous page**.

c) Is the solution obtained in part b) of this question optimal? Motivate your answer.

d) Write the dual of the given problem.

e) What can be said about the optimal objective function value of the dual, based on the solution found on part b)? Motivate your answer by precisely referring to the corresponding theoretical result.

Note: Part f) below (marked with **Quiz substitute**) replaces the score obtained in the quizzes. Answering it implies neglecting that score.

f) **Quiz substitute** On the **back of the previous page**, state the complementarity slackness conditions for the above pair of primal and dual problems and find a solution of the dual problem. Is it optimal? Motivate your answer.

Problem 3

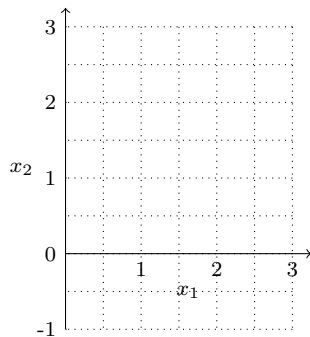
Part I: Consider the following Integer Linear Programming (ILP) problem:

$$\begin{array}{llllll} \max & z = & 5x_1 & +2x_2 & & \\ \text{s. t.} & & 2x_1 & +2x_2 & \leq & 5 \\ & & 2x_1 & -2x_2 & \leq & 3 \\ & & x_1, & x_2 & \geq & 0, \text{ integer} \end{array}$$

with the optimal tableau of the linear programming relaxation:

		x_1	x_2	x_3	x_4
	-11	0	0	$-7/4$	$-3/4$
x_1	2	1	0	$1/4$	$1/4$
x_2	$1/2$	0	1	$1/4$	$-1/4$

a) Draw the feasible region of the above ILP problem and report the optimal solution of its linear programming relaxation, plus the optimal objective function value.



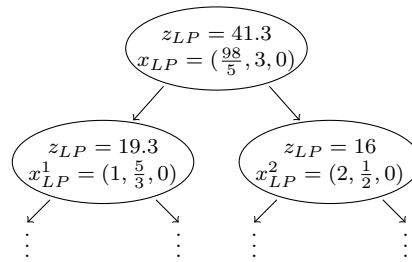
b) Determine the Gomory cutting plane associated with the basic variable x_2 . Is it the only one? Motivate your answer.

c) Express the above Gomory cutting plane in terms of the decision variables x_1 and x_2 and draw it on the graphical representation of the feasible region of the ILP problem presented in part a).

Part II (Please see the back of this page.)

Part II: Multiple choice question on branch-and-bound. Please mark the correct option(s). Wrong answers will be penalized.

Consider the following partial branch-and-bound tree of a maximization ILP problem.



1. Which is the lowest upper bound on the optimal objective function value of the original ILP?
 - i. 16
 - ii. 19.3
 - iii. 41.3
 - iv. None of the previous
2. Suppose that the subproblem (node) to be examined next is chosen according to the best-bound first strategy (with the best linear relaxation objective function value first). Then, which node should be examined first?
 - i. Node 1
 - ii. Node 2
 - iii. Either node 1 or node 2

Last name: _____

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Problem 4

Complete adequately the Python statements below to implement the following modeling components:

- integer parameters m, n
- sets $M = \{1, \dots, m\}$, $N = \{1, \dots, n\}$
- parameters π_{ij} , for any $i \in M, j \in N$
- parameters c_i , for any $i \in M$
- parameters p_j for any $j \in N$

```
import numpy as np
import mip
from mip import *
m = np.random.randint(100, size=1)[0] # this is an integer value
n = np.random.randint(100, size=1)[0] # this is an integer value
pi = np.random.rand(m, n)*100 # this numpy array represents an m x n matrix
c = np.random.rand(m)*90 # this is a vector with m components
p = np.random.rand(n)*10 # this is a vector with n components
```

- define the sets M, N

- declare the model

- nonnegative real variables $z_i, \forall i \in M$
- binary variables $y_j, \forall j \in N$

- the following objective function to be maximized:

$$\max \sum_{i \in M} c_i \left(z_i - \sum_{j \in N} y_j \right)$$

- the following constraints:

$$z_i \leq \sum_{j \in N: i \neq j} \pi_{ij} y_j \quad \forall i \in M$$

- the following constraints:

$$p_j y_j \leq z_i \quad \forall j \in N, i \in M \setminus \left\{ \frac{m+1}{2} \right\}$$

- optimize the model:

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Problem 5

An airline company is planning to rent a fleet of passenger aircrafts to cover a set of new long (LR), medium (MR) and short (SR) range routes during the next year. The one year rental price for each aircraft of type LR, MR, and SR is p_1 , p_2 and p_3 million euro, respectively. The company's direction board authorized a maximum budget of B million euro for renting the new planes.

All the rented aircrafts will be used to their maximum capacities. Therefore, the net profit estimated for each plane is r_1 , r_2 and r_3 million euro for planes of the LR, MR and SR types, respectively.

The airline company cannot rent more than 30 new planes. Moreover, if the number of rented SR and MR aircrafts is at least 20, then no more than 7 LR aircrafts can be rented.

If only SR planes were rented, the maintenance departments could operate up to 40 of these new planes. However, in terms of the use of these departments, the requirement for each MR plane is equivalent to $11/3$ of that for each SR plane and the requirement for each LR plane is equivalent to 4 times that for each MR plane.

Give an integer linear programming model formulation for the problem of deciding how many aircrafts of each type to be rented in order to maximize the total annual profit.

Parameters:

Decision variables:

Objective function:

Constraints: